

Rigorous Analysis: Bounding Box Algorithm Worst-Case Construction

1 Recursive Worst-Case Instance

Definition 1 (Recursive Point Set). *Let $n = 2^k$ for some integer $k \geq 1$. We define a point set $P_n \subset \mathbb{R}^2$ recursively as follows:*

- **Base case:** $P_2 = \{(0, 0), (1, 0)\}$
- **Recursive step:** For $n = 2^j$, define:

$$P_n = T_L \cup T_R$$

where:

$$\begin{aligned} T_L &= P_{n/2} \\ T_R &= \{(x + D_n, y) : (x, y) \in P_{n/2}\} \\ D_n &= 4n \end{aligned}$$

Geometric Properties

- All points lie along the line $y = 0$
- Total number of points: $|P_n| = n$
- Each recursive cluster $P_{n/2}$ lies in a contiguous horizontal segment of unit height and width $\approx n/2$
- At level j , the width of the entire point set is:

$$\text{width}(P_n) = D_n + \text{diameter}(P_{n/2}) = \Theta(n)$$

2 Behavior of the Bounding Box Algorithm

We analyze the standard bounding box divide-and-conquer heuristic:

1. Compute the bounding box of the point set
2. If width $>$ height, split with a vertical line; otherwise, split horizontally
3. Recursively solve left/right (or top/bottom) subproblems
4. Connect the two paths by choosing the shortest among the four possible endpoint-to-endpoint connections

Lemma 1 (Splitting Direction). *At every level, the algorithm splits the instance P_n with a vertical line, separating T_L and T_R .*

Proof. Since all points lie on $y = 0$, the bounding box has height 0 and width $D_n + O(n) = \Theta(n)$. Thus, the algorithm always chooses a vertical split. $\square$

Lemma 2 (Length of Connecting Edge). *The minimum distance between any point in T_L and any point in T_R is exactly $D_n = 4n$.*

Proof. All x -coordinates in T_L lie in $[0, O(n)]$, while all x -coordinates in T_R lie in $[4n, 4n + O(n)]$. Therefore, the shortest possible connection spans $D_n = 4n$ units. $\square$

3 Cost of the Algorithm's Output

Let $W_A(n)$ denote the total Wiener index of the path produced by the bounding box algorithm on P_n . We define the recurrence:

$$W_A(n) = 2W_A(n/2) + C(n), \quad W_A(2) = 1$$

where $C(n)$ is the contribution to the Wiener index from the connection between T_L and T_R .

Lemma 3 (Connection Cost).

$$C(n) = 4n \cdot \left(\frac{n}{2}\right) \left(\frac{n}{2}\right) = n^3$$

Proof. • The connecting edge has length $4n$

- In the final Hamiltonian path, this edge joins two subpaths of size $n/2$
- Its contribution to the Wiener index is:

$$\text{length} \times \text{position weight} = 4n \cdot \frac{n}{2} \cdot \frac{n}{2} = n^3$$

□

Theorem 1 (Bounding Box Algorithm Cost).

$$W_A(n) = \Theta(n^3 \log n)$$

Proof. We have the recurrence:

$$W_A(n) = 2W_A(n/2) + n^3$$

Apply the Master Theorem:

- $a = 2, b = 2, f(n) = n^3$
- $n^{\log_b a} = n$
- Since $f(n) = \omega(n^{\log_b a})$ and the regularity condition holds:

$$af(n/b) = 2 \cdot (n/2)^3 = \frac{n^3}{4} < cf(n) \text{ for } c = \frac{1}{2} < 1$$

The recurrence falls under Case 3 of the Master Theorem. Therefore:

$$W_A(n) = \Theta(n^3 \log n)$$

□

4 Optimal Solution Analysis

Let $W^*(n)$ denote the Wiener index of the optimal Hamiltonian path on P_n .

Theorem 2 (Optimal Cost).

$$W^*(n) = \Theta(n^3)$$

Proof. Construct the path π^* that visits:

- All points in T_L in increasing x -order
- A single jump from the rightmost point in T_L to the leftmost point in T_R
- All points in T_R in increasing x -order
- All intra-cluster edges are of length $O(1)$ with total weight $O(n^2)$
- The single inter-cluster edge has length $4n$ and weight $\left(\frac{n}{2}\right)^2$

Therefore:

$$W^*(n) = 2 \cdot O(n) \cdot O(n^2) + 4n \cdot \left(\frac{n}{2}\right)^2 = O(n^3) + n^3 = \Theta(n^3)$$

□

5 Approximation Ratio

Theorem 3 (Bounding Box Approximation Factor).

$$\frac{W_A(n)}{W^*(n)} = \Theta(\log n)$$

Proof. From Theorems 1 and 2:

$$\frac{W_A(n)}{W^*(n)} = \frac{\Theta(n^3 \log n)}{\Theta(n^3)} = \Theta(\log n)$$

□

Corollary (Unbounded Approximation Ratio). *The bounding box algorithm does not admit a constant-factor approximation for the Hamiltonian path Wiener index problem in the Euclidean plane.*